

Sales Data Cleaning Project Report

This report documents the data cleaning process performed on a sales dataset containing transactional information related to products, referral sources, payment methods, coupon usage, customer purchases, and revenue performance.

1. Project Objective

The objective of this project was to clean and prepare a raw sales dataset for accurate analysis and reporting. The dataset was standardized and validated to support future business intelligence and analytical processes.

2. Dataset Description

The dataset contains sales transaction records including Order IDs, Dates, Customer IDs, Products, Quantities, Unit Prices, Payment Methods, Coupon Codes, Referral Sources, Order Statuses, and Revenue information.

3. Cleaning Techniques Used

- Removed duplicate transaction records
- Corrected inconsistent date formats
- Standardized column data types
- Replaced missing/null values
- Corrected formatting inconsistencies
- Validated quantity and revenue fields
- Improved overall dataset consistency and structure

4. Missing Value Handling

Missing values were identified in the CouponCode column and replaced appropriately to improve dataset completeness and analytical reliability.

5. Duplicate Removal

Duplicate records were identified and removed to prevent inaccurate calculations, duplicate transactions, and misleading analytical results.

6. Formatting Corrections

- Standardized date formatting
- Corrected currency and numeric fields
- Updated text formatting consistency
- Verified proper column data types
- Ensured values were analysis-ready

7. Data Validation

The cleaned dataset was reviewed to confirm that duplicate records were removed, numeric columns contained valid values, revenue calculations were accurate, missing values were resolved, and formatting standards were consistently applied.

8. Result

The dataset was successfully transformed into a clean, structured, and analysis-ready format suitable for Exploratory Data Analysis (EDA), SQL querying, Power BI dashboard development, and business intelligence reporting.

9. Tools Used

- Microsoft Excel

10. Conclusion

This project demonstrated practical data cleaning techniques commonly used in real-world data analytics workflows. Proper data preparation improved data accuracy, consistency, and reliability, enabling the dataset to support meaningful analysis and business decision-making.